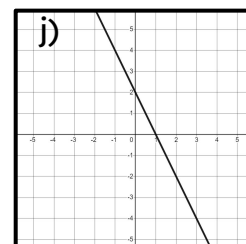
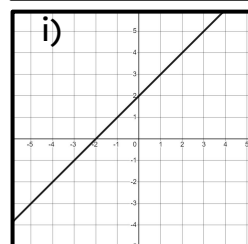
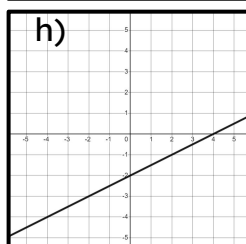
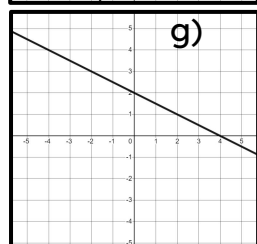
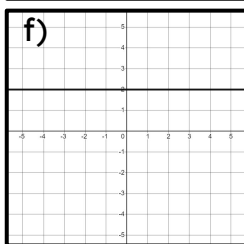
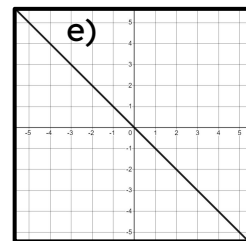
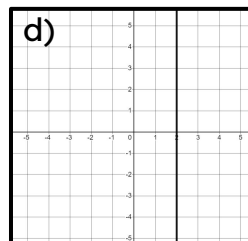
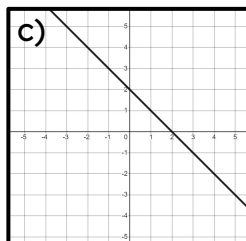
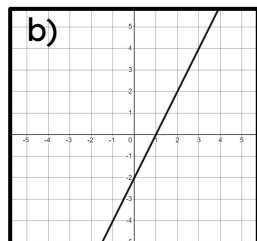
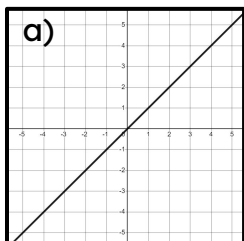




Finding the equation of the line $ay + bx + c = 0$

Task A: Equations in the form $y = mx + c$

1) Match each graph to its equation



$x = 2$

$y = 2$

$y = x$

$y = -x$

$y = 2 - \frac{1}{2}x$

$y = \frac{1}{2}x - 2$

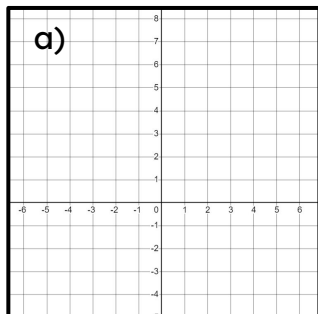
$y = 2x - 2$

$y = x + 2$

$y = 2 - x$

$y = 2 - 2x$

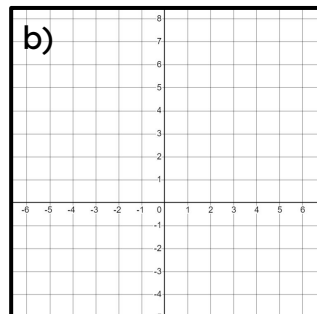
2) For each question, work out an equation for the line and use the key features to draw the line on the axes.



gradient: 3

y-intercept: (0, -2)

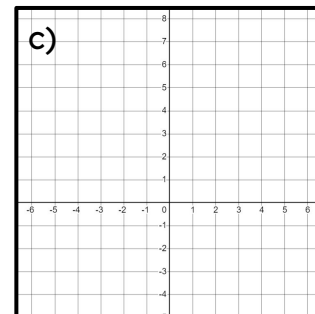
equation: _____



gradient: -1

y-intercept: (0, 5)

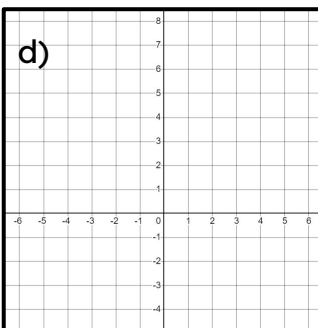
equation: _____



gradient: -4

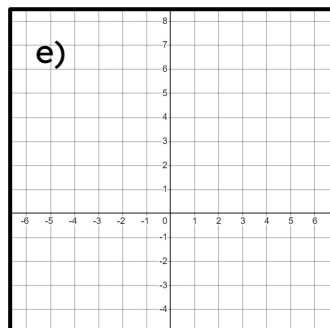
y-intercept: (0, -3)

equation: _____

gradient: $\frac{1}{2}$

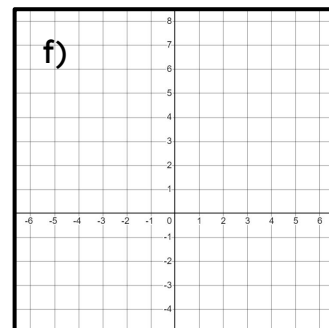
y-intercept: (0, 4)

equation: _____

gradient: $-\frac{1}{3}$

y-intercept: (0, -1)

equation: _____

gradient: $\frac{2}{3}$

y-intercept: (0, 0)

Equation: _____



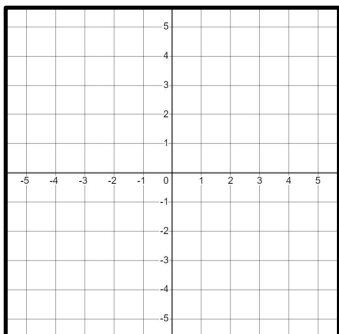
Task B: Equations in the form $ay + bx + c = 0$

1) For each equation, work out the y -intercept, the x -intercept and draw the graph.

a) $y + x - 4 = 0$

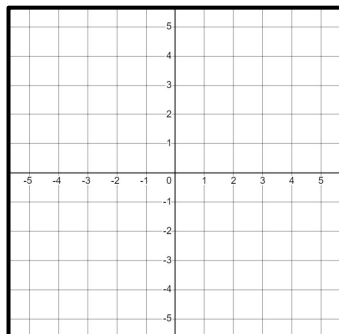
b) $y + x - 2 = 0$

c) $y + x + 2 = 0$



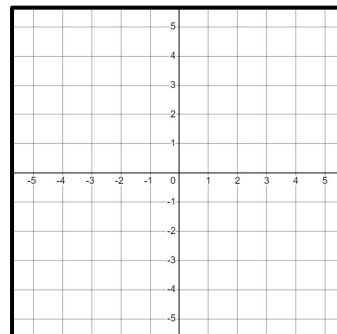
y -intercept (0, __)

x -intercept (__ , 0)



y -intercept (__ , __)

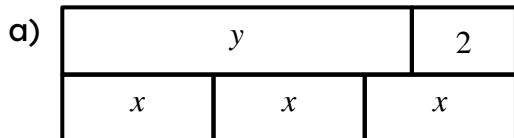
x -intercept (__ , __)



y -intercept (__ , __)

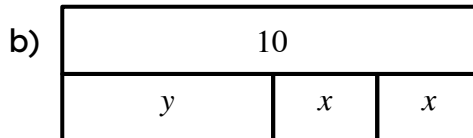
x -intercept (__ , __)

2) From the bar model, write the relationship as an equation in the form $y = mx + c$ and $ax + by + c = 0$



$y = mx + c$: _____

$ax + by + c = 0$: _____



$y = mx + c$: _____

$ax + by + c = 0$: _____

3) Fill in the missing working, then draw the graph of $2y + 5x - 20 = 0$

y -intercept

$2y + 5(0) - 20 = 0$

$2y - 20 = 0$

$2y =$ _____

$y = 10$ (__, __)

x -intercept

_____ + $5x - 20 = 0$

$5x -$ _____ = 0

$5x =$ _____

$x = 4$ (__, __)

